

CSE 220 SIGNALS AND SYSTEMS · FINAL EVALUATION

Sonar Distance Simulator

Pulse-Echo Ranging, Matched Filtering, and Range Resolution Limits

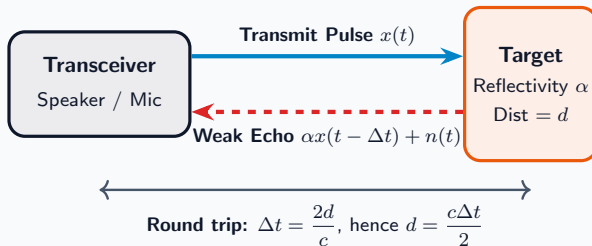
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Pure DSP Core · Monte Carlo Analysis · Streamlit Instrument UI · Real Hardware Ranging



1. Severe Noise

Echoes weaken as $1/r^2$.
At 0 dB SNR, the raw
waveform can hide them completely.

2. Echo Merging

Closely spaced targets create
overlapping echoes that appear
as one merged return.

3. Energy Dilemma

Short pulses resolve targets
but carry little energy; long pulses
are stronger but blur targets.

1. INPUTS

Pulse Controls:

- Waveform: Rect, Gauss, Chirp
- Duration T (e.g., 2 ms)
- Carrier f_c & Bandwidth B

Acoustic Channel:

- True Distances d_1, d_2
- Noise Level SNR_{dB}
- Speed $c = 343$ m/s

Configure

2. DSP ENGINE

Acoustic Channel

- Delay:
 $k = \text{round}(2df_s/c)$
- AWGN Noise Injection

Matched Filter

- $$r_{yx}[k] = \sum y[n]x[n-k]$$

Hilbert Envelope

- $e[n] = |\tilde{r}[n]|$, Peak Finder

Results

3. OUTPUTS

Ranging Numbers:

- Estimated Distance $\hat{d}$ (m)
- Absolute Error $|d - \hat{d}|$
- Multi-target list $[\hat{d}_1, \hat{d}_2]$

Visual Plots:

- Noisy Audio Buffer
- Sharp Correlation Spike

Audible Sound:

- Listenable .wav pings

Core Takeaway: Configure the pulse and scene, then receive distance estimates, diagnostic plots, and audible sonar clips.

Sliding Template Cross-Correlation:

- Mathematically optimal filter in AWGN (Cauchy-Schwarz Inequality):

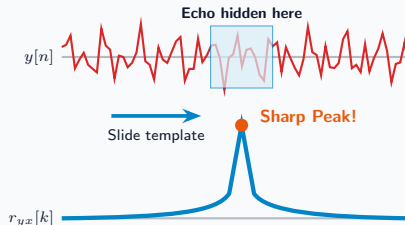
$$r_{yx}[k] = \sum_n y[n] x[n - k]$$

- Intuition:**

- Where there is only noise $\rightarrow$ multiplies random numbers $\rightarrow$ **averages to zero**.
- Where the echo matches the template $\rightarrow$ multiplies in-phase $\rightarrow$ **constructive surge!**

- Peak to Distance:**

$$\hat{d} = \frac{(k_{\text{peak}} - \text{offset}) \cdot c}{2f_s}$$



The Aha! Moment: Even when noise is louder than the echo (0 dB SNR), the matched filter recovers the echo with sub-millimeter precision!

1. The Carrier Trap (Raw Correlation):

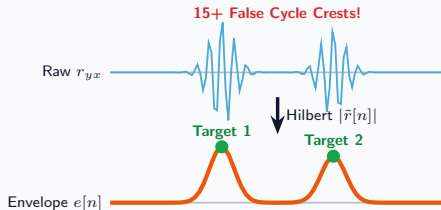
- Carrier oscillates at 4 kHz.
- Raw correlation produces **15–20 cycle ripples** for just 2 true targets!
- Simple thresholding counts false targets.

2. The Solution (Hilbert Envelope):

- Construct analytic signal:

$$\tilde{r}[n] = r[n] + j\mathcal{H}\{r[n]\}$$

- Smooth envelope: $e[n] = |\tilde{r}[n]|$.
- Peak finder on $e[n]$ detects **exactly 2 true target crests!**



Rule of Thumb: Raw argmax for single targets (sharpest); Hilbert envelope for multi-targets (eliminates carrier cycle ripples).

The Rayleigh Resolution Limit:

$$\Delta R_{\min} \approx \frac{c}{2B}$$

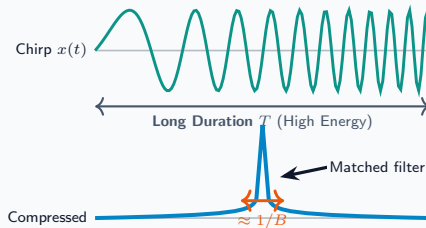
Two targets can only be separated if their echo gap exceeds the pulse's correlation width.

The Classical Trade-Off (Tone Bursts):

- Bandwidth is tied to duration: $B \approx 1/T$.
- To separate close targets $\rightarrow$ must make pulse tiny
 $\rightarrow$ carries tiny energy $\rightarrow$ ruined by noise!

The Chirp (LFM) Breakthrough:

- Sweeps frequency over time ($f_0 \rightarrow f_0 + B$).
- **Decouples duration T from bandwidth B :** high energy + razor-sharp resolution!



Pulse Compression: Long duration plus wide bandwidth gives high energy and a narrow $\approx 1/B$ correlation peak.

The Golden Rule

Functions in `src/` take numbers/arrays and return numbers/arrays. They NEVER call `plt.show()` or `print()`.

Why This Matters:

- **100% Reusable DSP Core:** Tests, report figures, and the Streamlit web app call the exact same functions.
- **Zero UI Coupling:** Front-end UI redesigns never risk breaking core physics calculations.
- **Fully Automated:** 33 passing pytest tests + 17 pulse physics verification checks.

`src/pulse.py` · Rect / Gaussian / Chirp generators; FFT -3 dB bandwidth



`src/channel.py` · Sample delay, geometric loss, AWGN injection



`src/receiver.py` · Matched filter, single-target & Hilbert multi-target

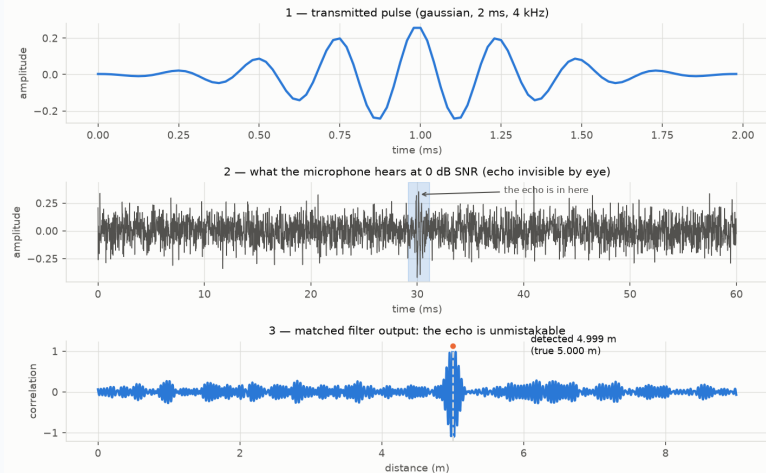


`src/evaluate.py` · Monte Carlo noise & resolution sweeps



Shared Consumers

tests/ (33 tests) | notebooks/
| app/streamlit_app.py



1. Transmit:

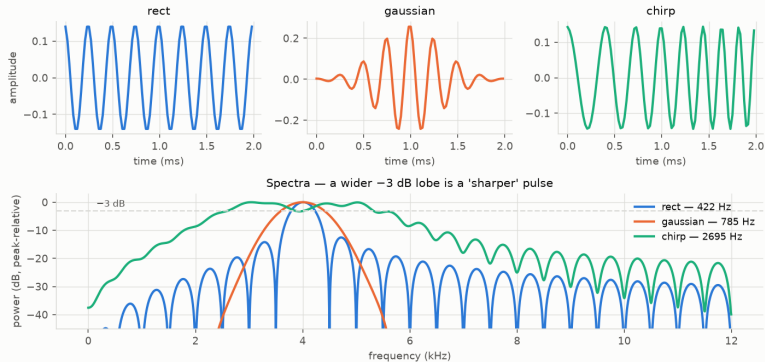
Gaussian pulse (2 ms, 4 kHz) normalized to unit energy.

2. Corrupted:

At 0 dB SNR, echo is completely drowned by ambient noise!

3. Recovered:

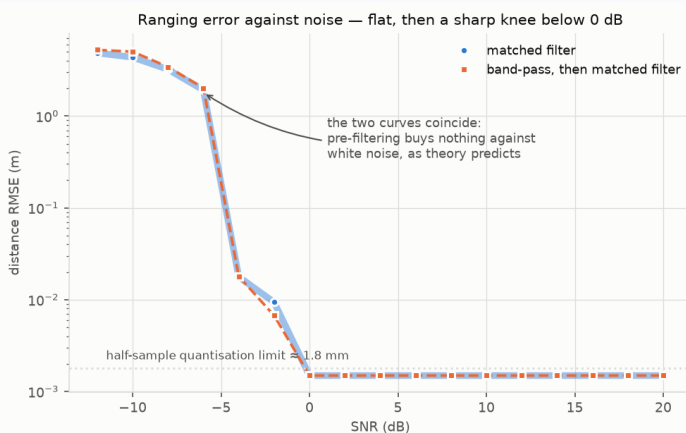
Matched filter yields a towering peak at **5.000 m** (< 1 mm error!).



Waveform Comparison:

- **Rect:** Abrupt on/off switch → heavy spectral splatter.
- **Gaussian:** Bell envelope → smooth, clean spectrum.
- **Chirp:** Linear frequency sweep → wide -3 dB bandwidth for sharp target separation.

Key Metric: A wider -3 dB spectral lobe yields a sharper correlation peak.



Key Observations:

1. Quantization Floor (≈ 1.8 mm):

- Above 0 dB, error is flat.
- Arises from discrete sampling:

$$\text{Floor} \approx \frac{c}{4f_s} \approx 1.79 \text{ mm}$$

2. The Threshold Cliff:

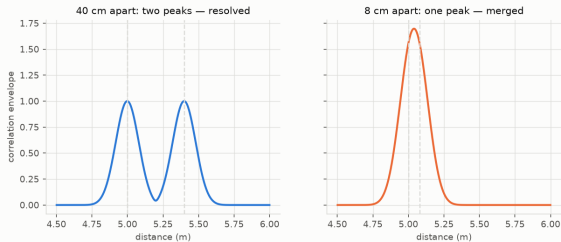
- Below -4 dB, noise spikes eclipse the echo peak, causing rapid breakdown.

3. Bandpass Filter Overlap:

- Pre-filtering coincides with raw matched filter $\rightarrow$ confirms matched filter is already optimal for white noise!

Multipath Separation (40 cm vs. 8 cm)

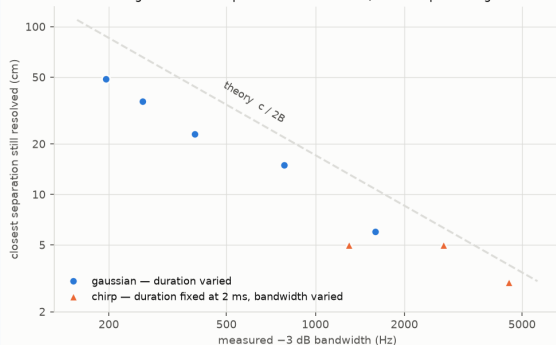
Two targets, same pulse — the limit is about 15 cm here



- 40 cm gap: Two distinct peaks resolved.
- 8 cm gap: Echoes merge into a single wide blob.

Measured Resolution vs. Bandwidth

Range resolution improves with bandwidth, not with pulse length

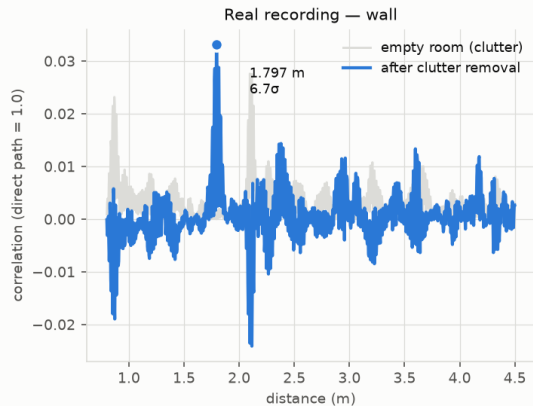


- Measured points track theoretical dashed curve $c/(2B)$ across a $23\times$ bandwidth range.
- Chirps fix duration at 2 ms while sweeping bandwidth.

4-Step Hardware Acoustic Pipeline:

- 1 **Windowed Chirp:** 2–8 kHz sweep shaped to laptop speaker response.
- 2 **Direct-Path Sync:** Lock onto speaker-to-mic direct bleed as $t = 0$ origin.
- 3 **Frame Folding:** Coherently average 4 bursts $\rightarrow$ cancels room noise.
- 4 **Clutter Removal:** Subtract empty-room baseline $\rightarrow$ removes furniture reflections and speaker ring.

Target	Truth	Measured	Quality
Room Wall	≈ 1.80 m	1.797 m	6.7σ
Raised Hand	≈ 0.30 m	0.272 m	6.2σ



Statistical Confidence: Both detections exceed $> 6.0\sigma$ above background clutter! Bit-for-bit verified in `tests/test_acoustic.py`.

Four Dedicated Operating Modes:

- 1 **01: Single Target Ranging**
 - Real-time sliders for distance, SNR, and pulse shape.
- 2 **02: Multi-Target Resolution**
 - Live separation slider and dynamic bandwidth sweep.
- 3 **03: Noise Resilience Benchmark**
 - Automated Monte Carlo SNR sweep with progress bar.
- 4 **04: Real Acoustic Hardware Test**
 - Live and archived laptop microphone/speaker ranging.

Listenable Audio Engine (src/audio.py)

- **4× Slowdown:** Declares lower playback rate → stretches 2 ms pulse into audible sound with zero resampling distortion.
- **Gap Pacing:** Inserts silence gaps for realistic "ping... ping" cadence.
- **Click-Free Loops:** Faded edges eliminate audio boundary clicks.
- **Independent Noise:** Fresh random seeds per frame avoid artificial cyclic hissing.

1. Baseline Single-Target (Menu 1)

Set $d = 5.0$ m, SNR = 10 dB. Show clean correlation peak and play the audible ping.

2. The Invisible Echo at 0 dB (Menu 1)

Drop SNR to 0 dB. Show that echo is completely invisible in time domain, yet matched filter gets 4.999 m!

3. Multi-Target Collision (Menu 2)

Drag Target 2 from 40 cm gap to 8 cm gap. Watch the two separate peaks merge into one broad blob!

4. The Chirp Resolution Fix (Menu 2)

Switch from Gaussian to **Chirp** ($B = 4$ kHz). Instantly, the single merged blob splits back into two distinct peaks!

Hardware Backup: Switch to Menu 4 to showcase the physical wall detection at 1.797 m (6.7σ).

Aditya Tirtho Roy

Student ID: 2305178

Core DSP & Mathematical Modeling:

- Mathematical formulation of matched filtering, envelope detection, and range resolution.
- Implemented core DSP modules in `src/pulse.py`, `channel.py`, `receiver.py`, `filters.py`, `evaluate.py`.
- Built unit test suite (33 tests) and pulse validation (`check_pulse.py`).
- Generated Monte Carlo report figures.
- Hardware DSP pipeline (`src/acoustic.py`).

Farhan Ehsas Sami

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Application, Audio & Physical Testing:

- Architecture and implementation of the complete Streamlit UI (`app/streamlit_app.py`).
- Designed Plotly visualization stages and dark-mode instrument styling.
- Built listenable audio engine (`src/audio.py`): 4× slowdown, gap pacing, and loop fading.
- Physical laptop acoustic experimentation and data capture (`record_acoustic.py`).
- System integration and documentation.

1. Matched Filter

Provably optimal detector in AWGN. Recovers signals buried at 0 dB SNR with sub-millimeter accuracy. Pre-filtering adds no gain against white noise.

2. Chirp Compression

Frequency modulation decouples energy from resolution. Wide -3 dB bandwidth separates close targets without sacrificing range.

3. Reality Verified

Closed the loop from simulated arrays to physical laptop acoustic ranging with $> 6\sigma$ confidence and 100% test coverage.

Thank You!

Questions and Discussion Welcomed

Q1: Why does RMSE plateau at ≈ 1.8 mm instead of 0?

- **Answer:** Time delay is quantized into integer samples: $k = \text{round}(2df_s/c)$. Maximum rounding error is ± 0.5 samples. At $f_s = 48$ kHz, half a sample is:

$$\Delta d_{\text{quant}} = \frac{c}{4f_s} = \frac{343}{4 \times 48000} \approx 1.79 \text{ mm}$$

Q2: Why did Butterworth bandpass pre-filter fail to improve RMSE?

- **Answer:** The matched filter transfer function $H(f) = X^*(f)$ already weights each frequency by its signal amplitude. By Cauchy-Schwarz, it is mathematically optimal for AWGN. Pre-filtering out-of-band noise is redundant, while filtering in-band risks attenuating signal power.

Q3: Why raw argmax for single target, but Hilbert for multi-target?

- **Answer:** Single-target ranging achieves highest precision with the sharpest peak (raw correlation argmax). But for multi-targets, raw cross-correlation oscillates at the 4 kHz carrier, creating 15+ false cycle crests. The Hilbert envelope $|\tilde{r}[n]|$ eliminates carrier ripple to reliably count true targets.

Q4: How did you cancel direct speaker bleed on hardware?

- **Answer:** Speaker-to-mic bleed arrives first (≈ 10 cm) and synchronizes $t = 0$. We then coherently fold 4 frames and subtract an empty-room baseline recording (clutter removal), revealing reflections above 6σ .